Measures of Macroeconomic Uncertainty: I3E Index and Its Alternatives

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Abstract

The quantification of economic uncertainty has garnered significant scholarly interest over the past decade. A range of alternative uncertainty measures has emerged in the literature, each characterized by distinct methodological approaches and scope of applicability. Among the most recent contributions is the I3E Index of Economic Uncertainty. We assess the validity of the I3E index as a proxy for economic uncertainty. Furthermore, we benchmark its performance against some widely utilized measures of economic uncertainty in academic research: Economic Policy Uncertainty Index (EPU), World Uncertainty Index (WUI), Geopolitical Risk (GPR), and Volatility Index (VIX).

Keywords: Macroeconomic Uncertainty, I3E Index, EPU, WUI, VIX.

1. Introduction

A sequence of major global shocks—including the 2008 financial crisis, Brexit, the COVID-19 pandemic, Ukraine, the resurgence of trade conflicts in 2025, and the recent conflict involving Iran in 2026—has underscored the central role of economic uncertainty as a key driver of macroeconomic dynamics.

Uncertainty may not be directly observable, but it can be proxied through measurable indicators. A number of uncertainty indices and empirical approaches have emerged in recent years, each tailored to specific data sources, country coverage, or methodological underpinnings (Cascaldi-Garcia et al., 2023). The literature broadly distinguishes between micro and macro uncertainty (Altig et al., 2020; Bachmann & Bayer, 2013, 2014; Bachmann et al., 2013; Castelnuovo, 2019; Dibiasi & Sarferaz, 2023; Fernández-Villaverde et al., 2020), survey-based (Altig et al., 2022; Leduc & Liu, 2016; Scotti, 2016), text-based (Ahir et al., 2022; Baker et al., 2016; Caldara & Iacoviello, 2022; Husted et al., 2020) and volatility-based uncertainty (Andersen et al., 2003; Bloom, 2009; Fernández-Villaverde et al., 2011; Jurado et al., 2015; Leahy & Whited, 1996; Lensink et al., 1999; Ludvigson et al., 2021).

Micro and cross-sectional dispersion indices seek to capture uncertainty by measuring the level of disagreement among firm profits in an industry or among forecasters, managers and other economic agents regarding sales, profits, consumption, and other related economic variables (Altig et al., 2020; Bachmann & Bayer, 2013). Some of these indices use survey data and tend to be more idiosyncratic to a particular country, sector, firm, or time period, making the comparison across countries or time less straightforward.

Given our focus on comparative macroeconomic uncertainty over long periods of time, we consider only text-based measures and volatility indices in this research. In particular, we compare text (WUI, EPU, GPR) and volatility (I3E, VIX) uncertainty indices.

Text-based indices analyze the frequency of words related to uncertainty within a large corpus of text, such as newspapers, policy documents, financial reports, or social media. These indices aim to quantify economic uncertainty based on the idea that the more economic agents talk about uncertainty, the more uncertain the environment is. Examples include the World Uncertainty Index (WUI) (Ahir et al., 2022), Economic Policy

Uncertainty (EPU) (Baker et al., 2016), Geopolitical Risk Index (Caldara & Iacoviello, 2022) and the web-search measure EURQ (Bontempi et al., 2021).

Volatility-based measures rest on the idea that uncertainty is reflected in the instability of key economic and financial variables. When economic agents face a more uncertain environment, expectations about future outcomes become more dispersed and are revised more frequently, which tends to generate greater variability in asset prices, interest rates, exchange rates, and other macro-financial indicators.

Volatility-based indices are typically built using the annual, quarterly, monthly, daily or even real-time variability of a number of key economic and financial variables (Bloom, 2009; Fernández-Villaverde et al., 2011; Jurado et al., 2015; Leahy & Whited, 1996; Lensink et al., 1999; Ludvigson et al., 2021). These indices may use normal statistical variance, the variance of the unpredictable part of a stochastic process, the conditional variance estimated from General AutoRegressive Conditional Heteroskedastic (GARCH) models or variance estimates from the geometric Brownian motion.

Some volatility indices measure anticipated or expected economic uncertainty. For instance, the VIX, often labeled the "fear index", measures the *expected* 30-day volatility implied by stock index option prices like the S&P500. The VIX is published by the Chicago Board Options Exchange (CBOE) since 1990 and is one frequently used proxy for financial uncertainty (Whaley, 1993, 2009).

The I3E economic uncertainty index (Garcia-Castro & Ariño, 2026) is a general volatility-based index providing a measure of economic uncertainty using the daily volatility of macroeconomic indicators available for most developed countries. The I3E Spot, or simply I3E, is computed using the realized values of the series whereas the I3E Forward is computed using the option-implied volatility of the series. We consider both in this article.

The remainder of the paper is organized as follows. Section 2 reviews the measures of macroeconomic uncertainty considered in this study, together with their methodologies and underlying data. Section 3 compares the evolution of the indices over the last decades and examines their comovements. Sections 4 and 5 explore how variants of the I3E relate to established measures of macrouncertainty. The paper concludes with a final section.

2. Measures of Macroeconomic Uncertainty

Table 1 summarizes measures of economic uncertainty commonly employed in empirical studies. We review next the indices compared in this study.

2.1 Economic Policy Uncertainty (EPU)

The EPU is a text-based index based on the frequency of a trio of terms related to the economy (E), policy (P) and uncertainty (U) in a selection of national newspapers (Baker et al., 2016). For instance, to compute the EPU index for the United States the following 10 leading newspapers are used: USA Today, the Miami Herald, the Chicago Tribune, the Washington Post, the Los Angeles Times, the Boston Globe, the San Francisco Chronicle, the Dallas Morning News, the Houston Chronicle, and the WSJ.

For each newspaper, it counts the term 'uncertainty' or 'uncertain', 'economic' or 'economy', and one or more of the following policy terms: 'congress', 'legislation', 'white house', 'regulation', 'federal reserve', or 'deficit'. Then the raw count of policy uncertainty articles are divided by the total number of articles in the same paper and month. The resulting series for each paper are then normalized to have a unit standard deviation from January 1985 through December 2009. Next, all normalized values over papers in each month are summed to obtain a multi-paper index. Finally, the multi-paper index is re-normalized to an average value of 101.8 from January 1985 through December 2009.

The same methodology is followed in other advanced economies to produce national EPU indices using the leading newspapers in each country. EPU data and methods as well as the list of countries being covered are available here https://www.policyuncertainty.com/index.html.

2.2 World Uncertainty Index (WUI)

The WUI has emerged as one of the most widely used text-based measures of uncertainty in recent years (Ahir et al., 2022). It uses the frequency of the term "uncertainty" (and its variants) in country-specific reports produced quarterly by the Economist Intelligence Unit (EIU). Specifically, the index is computed as follows:

$$\text{WUI}_t = \left(\frac{\text{Number of mentions of uncertainty and related terms}}{\text{Total number of words in the report}} \right) \times 1,000 \qquad (1)$$

Thus, it measures the occurrence of the term "uncertainty" per thousand words, providing a normalized indicator suitable for cross-country comparisons. The WUI is available quarterly for 143 countries, extending back as far as 1952. The WUI is also available monthly for 71 countries from 2008 to the present.

WUI data and full methodology can be obtained from https://worlduncertaintyindex.com.

2.3 Geopolitical Risk Index (GPR)

The Geopolitical Risk (GPR) Index developed by Caldara and Iacoviello (2022) is a text-based measure designed to capture geopolitical tensions at the global level. The index is constructed from a systematic analysis of newspaper articles, focusing on terms related to geopolitical tensions, such as wars, military conflicts, terrorist acts, and diplomatic crises.

Specifically, the GPR is computed by counting the number of articles that contain a set of predefined keywords associated with geopolitical risk in a group of leading international newspapers. These counts are then scaled by the total number of articles in each newspaper and period, yielding a normalized measure comparable over time.

The GPR index is available at both monthly and daily frequencies, with the historical series extending back to the early 20th century for several countries. The index has been widely used in empirical research to study geopolitical shocks, including their impact on investment, asset prices, and economic activity.

GPR data and detailed methodology are available at https://www.matteoiacoviello.com/gpr.htm.

2.4 CBOE Volatility Index (VIX)

The VIX is a forward-looking measure of financial uncertainty. It measures volatility investors expect to see (Whaley, 1993, 2009). The value of the VIX is implied by the current prices of options on the S&P 500 Index and represents expected future stock

market volatility over the next 30 calendar days, annualized.

This index is calculated in real time by the Chicago Board Options Exchange (CBOE) using a range of SPX options. It is constructed using the prices of a portfolio of out-of-the-money (OTM) SPX call and put options. Rather than assuming a specific model of volatility (e.g., GARCH or Black-Scholes implied volatility), the VIX reflects the full risk-neutral distribution as inferred from option prices.

The VIX is computed using the following formula:

$$\sigma^2 = \frac{2}{T} \sum_i \frac{\Delta K_i}{K_i^2} e^{RT} Q(K_i) - \frac{1}{T} \left(\frac{F}{K_0} - 1 \right)^2 \tag{2}$$

$$\text{VIX} = 100 \times \sqrt{\sigma^2} \tag{3}$$

where σ^2 is the 30-day variance, and the terms are defined as follows:

- T is the time to expiration in years
- K_i is the strike price of the i-th out-of-the-money option.
- ΔK_i is the interval between strike prices: $\Delta K_i = \frac{K_{i+1} - K_{i-1}}{2}$.
- $Q(K_i)$ is the average of the bid and ask prices of the OTM option with strike K_i.
- R is the risk-free interest rate corresponding to the time to expiration.
- F is the S&P 500 forward price from put-call parity near-the-money options.
- K_0 is the strike price closest to F (the forward price).

The VIX calculation uses a weighted average of two SPX options series with maturities that bracket the 30-day horizon. The series are then interpolated to obtain a 30-day constant maturity estimate of expected volatility. The final VIX value is quoted as an annualized standard deviation in percentage points (e.g., a VIX of 20 implies a 20% annualized implied volatility).

A higher VIX indicates that investors expect greater volatility in the near term, often due to macroeconomic uncertainty, geopolitical risk, or earnings announcements. Conversely, a low VIX suggests market complacency or confidence.

The VIX has been extended beyond the S&P 500, both across asset classes and across countries. There are VIX-style implied-volatility indices for other equity benchmarks and non-equity assets, including the Nasdaq-100, Russell 2000, crude oil (OVX), gold (GVZ), the euro exchange rate (EVZ), and U.S. Treasuries (VIXTLT), all derived from option prices. In Europe, the VSTOXX measures 30-day implied volatility from EURO STOXX, and the VIBEX the same for the IBEX-35. In Asia, the Japan Exchange Group publishes volatility indexes linked to Nikkei options, and the National Stock Exchange of India publishes the India VIX based on NIFTY option prices.

2.5 Economic Uncertainty Index (I3E)

The I3E economic uncertainty index (Garcia-Castro & Ariño, 2026) is a composite volatility-based index constructed from daily movements in four macro-financial variables: equity prices, 10-year government bond yields, exchange rates, and Brent oil prices. It combines the log-transformed, exponentially smoothed volatilities of these series into a standardized indicator with mean 100, thereby allowing consistent comparisons across countries and over time. It fluctuates between a theoretical value of 0 and 200 points. The I3E is published daily and currently covers 24 countries.

The I3E Global is defined as the simple average of the national I3E indices, thereby assigning equal weight to each country regardless of economic size. Alternatively, a GDP-weighted version of the I3E Global is also available, in which each country's contribution is proportional to its nominal GDP in U.S. dollars.

The I3E Forward is constructed analogously to the I3E Spot, except that it relies on one-month constant-maturity option-implied volatilities for the same four underlying series. These implied volatilities are log-transformed, standardized, and aggregated using the same normalization procedure as in the spot index. The resulting measure captures forward-looking macro-financial uncertainty as reflected in market expectations of volatility over the next month.

Comprehensive technical details on the construction of the I3E, I3E Global, and I3E Forward indexes, along with the complete historical daily dataset is available at `i3e-index.github.io`.

3. Comparative Analysis

We compare the EPU, WUI, GPR, VIX, and I3E indices. For all measures except the VIX, we use their global values. The VIX corresponds to the implied volatility of options on the S&P 500, a global reference for stock market volatility.

Figure 1 compares I3E with EPU. The two series display broadly similar medium-run movements, with both rising during periods of heightened global stress. However, their short-run dynamics differ noticeably. This suggests that the two indicators share a common uncertainty dimension, but differ in amplitude and timing, with EPU more responsive to acute policy-related events. Following the U.S. election on November 2024, the EPU presents a noticeable increase, several orders of magnitude larger than the I3E. The perception of uncertainty in the media during that period was higher than the realized volatility in the market. Following the U.S. attack on Iran, the I3E rose rapidly, closing the gap with the EPU. Before 2024, the EPU remained below I3E for extensive periods.

Figure 2 compares I3E with WUI. The correlation is positive until 2024, after which it is no longer significant. Both measures tend to rise during episodes of heightened global uncertainty, particularly from the late 2000s onward, which suggests that they capture a common underlying uncertainty component. WUI remains relatively subdued through much of the 1990s and early 2000s, then trends upward more gradually before rising sharply at the end of the sample, whereas I3E displays much larger and more frequent short-run fluctuations throughout the period. The pattern suggests that the two indicators are related, but differ in timing, amplitude, and likely in the dimension of uncertainty they capture.

Figure 3 compares I3E and GPR. GPR mainly reacts to major geopolitical global events such as the Iran conflict in 2026 and global trade disputes in 2025 but remains low during periods of financial stress like 2008. Thus, the correlation with I3E is relatively low. However, as shown below, GPR shows a higher correlation with the I3E subindices more closely related to geoeconomics, like exchange rates and oil prices.

Figure 4 compares I3E with VIX. As expected, the two series exhibit the most pronounced comovement, particularly during major episodes of global stress. However, their short-run dynamics differ. The VIX reacts with sharp and short-lived spikes, whereas

the I3E evolves more smoothly and remains elevated for longer periods. This pattern appears to reflect the fact that the VIX primarily reflects high-frequency market-based uncertainty, while the I3E captures a broader and more persistent uncertainty process, some of which are political and geoeconomic in nature, and not necessarily financial.

Pairwise correlations across indices are shown in Table 2. For comparability, daily indices (I3E, WIX, GPR) are converted to monthly frequency by taking monthly averages.

Among the text-based indices, EPU and WUI exhibit the strongest correlation (0.641), while GPR is less closely related. We find a positive correlation of I3E with EPU (0.159) and with GPR (0.104). The negative correlation between WUI and I3E is not significant. The absence of strong correlations between text-based and volatility-based uncertainty metrics is in line with previous findings (Ahir et al., 2022). Such disconnect has widened over time, particularly since 2024, with a rapid increase of uncertainty counts in major newspapers without a commensurate rise in either I3E or VIX. Over the period 1990-2024, the correlation between WUI and I3E (0.140) and between EPU and I3E (0.262) is positive and significant.

A recurrent pattern in the graphs is that WUI, EPU and GPR tend to spike during anticipated events such as trade disputes, war-related announcements, and pandemics but fail to do so during episodes of financial stress. One possible explanation is that global trade disputes are often signaled and discussed in advance in the media, whereas financial crisis by their nature, arrive unexpectedly. As argued by Kay and King (2020), the collapse of Lehman Brothers on 15 September 2008 could not have been predicted in advance, because if it had been, it would not have occurred in that form or at that time.

By contrast, the VIX and I3E capture the turbulence in the 2007-2008 period. The largest spikes occur first almost simultaneously in September 2008, and then again following the collapse of Lehman Brothers in October 2008.

Compared to VIX, the I3E correlation with text-based indexes is slightly higher, reflecting the global and more comprehensive nature of the I3E. The standard S&P 500-based VIX, while often used as a global benchmark for market stress, is inherently tied to U.S. and to financial volatility. Nevertheless, the VIX and the I3E display strong comovement over the sampled period, with a statistically significant correlation of 0.760.

To provide a more-grained analysis, the table also includes proxies for financial uncertainty ($I3E_{FIN}$), political uncertainty ($I3E_{POL}$), and geoeconomic uncertainty ($I3E_{GEO}$).

The first two are the I3E subindexes corresponding to stock market and government bond volatility respectively. The index $I3E_{GEO}$ is the simple average between the I3E exchange rate and oil Brent subindexes.

The breakdown reveals that $I3E_{FIN}$ is more strongly correlated with indexes related to financial volatility like the VIX (0.812). The $I3E_{POL}$ is strongly correlated with EPU (0.543), a political uncertainty index, and to a lesser extent with the WUI (0.121). Interestingly, $I3E_{GEO}$ exhibits the strongest correlation with GPR (0.147) among all indices, consistent with the multi-faceted nature of the I3E.

The results provide empirical support for our breakdown of uncertainty into financial, political and geoeconomic and allow to connect the I3E index —and the subindices $I3E_{FIN}$, $I3E_{POL}$, $I3E_{GEO}$— with the extensive empirical literature that have distinguished between financial (VIX, GARCH models, market volatility), economic policy (EPU) and geoeconomic (GPR) uncertainty metrics.

Overall, both text-based and volatility-based indices exhibit pronounced responses to some major economic shocks, such as the COVID-19 pandemic or the global trade dispute in 2025. Yet there are periods —such as 1996, 2014, 2025— where the trajectories of the indices diverge notably. These episodes of decoupling suggest that the indices may be capturing different dimensions or sources of economic uncertainty.

The VIX and the I3E follow similar paths, although the VIX tends to have a higher floor during periods of low uncertainty compared to the I3E. In times of heightened uncertainty, like in 2008 and 2020, the VIX rises to a higher ceiling than the I3E and the WUI. Throughout the analyzed period, the WUI exhibits less pronounced peaks and troughs than those observed in the I3E and VIX.

The EPU and WUI are derived from text-based analysis of country reports and newspapers, making them well suited for cross-country comparisons and long-run analyses. Their broad coverage across economies and over time provides a unique perspective on global and national policy uncertainty. They are most applicable in scenarios where researchers seek to study structural or persistent sources of uncertainty, such as the long-term effects of political transitions, trade disputes, or global crises.

By contrast, the VIX is derived from option-implied volatility in U.S. equity markets. It provides real-time, high-frequency information on financial market sentiment and is particularly sensitive to sudden shocks. Its main applicability is in scenarios charac-

terized by acute episodes of stress such as the 2008 global financial crisis where market expectations about risk and volatility shift rapidly.

Finally, the I3E index exhibits greater responsiveness than the WUI and EPU during episodes of financial stress. It is particularly well suited for analyzing the interaction between uncertainty and macroeconomic variables such as global investment, consumption, and financial markets. Owing to its construction, the I3E enables researchers to bridge the gap between qualitative, long-run measures and high-frequency market-based indicators. Its main strength lies in providing a comprehensive proxy for economic uncertainty in environments where financial, political, and geoeconomic factors may be closely intertwined.

4. Forward Indices

The I3E forward uses the option-implied one-month volatilities of the four underlying series to provide a measure of the uncertainty expected by economic agents over the next month (Garcia-Castro & Ariño, 2026). We compare the correlations between I3E, I3E forward, and alternative uncertainty metrics in Table 3.

Correlations between the I3E forward indices and text-based uncertainty measures (WUI and EPU) are generally lower than those obtained for the I3E Spot indices. By contrast, correlations with the GPR are marginally higher. Consistent with the fact that both the VIX and the I3E forward are constructed from one-month option-implied volatility, the U.S. I3E forward exhibits a relatively strong correlation with the VIX (0.787). At the same time, the correlation is well below unity, indicating that the two measures are not interchangeable and likely reflect differences in their underlying construction. In particular, whereas the VIX is derived solely from S&P 500 options, the I3E forward aggregates implied volatilities across a broader set of asset classes, including equity indices, government bond yields, exchange rates, and oil prices.

The divergence between the I3E Spot and I3E forward measures is consistent with evidence that realized and implied volatility are empirically distinct and have heterogeneous effects on economic growth (Berger et al., 2020; Yang, 2025).

Taken together, these results suggest that the I3E forward captures a dimension of uncertainty that is distinct from that reflected in the I3E Spot measure, and more closely

associated with forward-looking expectations embedded in option markets than with realized volatility.

5. Conditional-Variance Uncertainty

Previous research has emphasized that realized volatility need not coincide with economic uncertainty properly understood. Jurado et al. (2015), widely regarded as the benchmark measure of macrouncertainty for the United States, define uncertainty as the conditional volatility of the unforecastable component of future economic outcomes (U_{JLN}), rather than as the ex post realized volatility of the series itself. Related contributions, including Choi et al. (2015) and Fountas and Karanasos (2006), use econometric models such as GARCH to estimate the conditional second moment of innovations in financial or macroeconomic variables. The common insight in this literature is that "true" uncertainty should be interpreted as the ex ante conditional second moment of the unpredictable component of the variable of interest, that is, the component that cannot be inferred from the information set available at time t.

In a similar spirit, Garcia-Castro and Ariño (2026) exploit one-month option-implied volatility of the underlying series entering the I3E index to recover the conditional volatility of the unforecastable component of one-month-ahead I3E values, thereby isolating genuine uncertainty (U_{GCA}) from realized volatility.

Whereas U_{JLN} is estimated from a broad panel of macroeconomic indicators and is available only for the United States, U_{GCA} relies exclusively on the option-implied volatilities of the four constituent series underlying the I3E. Notwithstanding these differences in information sets and modelling assumptions, the two measures exhibit a high correlation.

Table 4 presents the results of the GARCH-based model, where current and lagged values of the I3E Forward indices for the United States, Europe, and Japan are used to estimate one-month-ahead values of the I3E Global Spot index.

A striking finding is that U_{GCA} is more strongly correlated with both EPU and WUI than the I3E Global Spot index itself, and that these correlations are positive and statistically significant (Table 5). This result is noteworthy, as prior research has documented only modest or even negligible correlations between WUI and other volatility-based measures of uncertainty (Ahir et al., 2022; Castelnuovo, 2019).

This suggests that our conditional-volatility-based measure isolates a dimension of uncertainty that is more closely aligned with established broad-based proxies of economic uncertainty than standard measures of realized volatility. In other words, filtering out the forecastable component appears to sharpen the informational content of the I3E with respect to underlying uncertainty.

The correlation between U_{GCA} and GPR is virtually the same as that between GPR and the plain I3E, and remains lower than the corresponding correlations with WUI and EPU. A plausible explanation is that GPR is concentrated on a limited number of major geopolitical episodes, such as September 11 and the war in Ukraine, and therefore captures sharp event-specific spikes rather than the broader and more persistent uncertainty reflected in EPU, WUI, and U_{GCA}. This suggests that U_{GCA} is not merely capturing salient geopolitical shocks, but rather a broader and more persistent dimension of macroeconomic uncertainty.

6. Conclusion

This paper examines the measurement of macroeconomic uncertainty, with particular attention to the recently developed I3E Index. Drawing on a review of leading uncertainty metrics such as the World Uncertainty Index (WUI), Economic Policy Uncertainty (EPU), Geopolitical Risk Index (GPR), and Volatility Index (VIX), the study situates the I3E as an alternative volatility-based index that synthesizes daily data from four key financial indicators: domestic stock index prices, government bond prices, exchange rates, and international oil prices. The I3E offers extensive country coverage and a daily frequency since 1990, providing a tool for researchers interested in comparative or historical analyses of economic uncertainty.

The comparative analysis highlights both commonalities and important differences across uncertainty measures. While all indices respond to major global shocks, their dynamics differ markedly in terms of timing, persistence, and amplitude (Jurado et al., 2015; Ludvigson et al., 2021). Text-based measures such as the EPU and WUI tend to capture slower-moving and more persistent perceptions of uncertainty (Ahir et al., 2022; Baker et al., 2016), whereas market-based indicators such as the VIX reflect high-frequency fluctuations in financial conditions (Bloom, 2009). The GPR appears to capture

acute geopolitical and geoeconomic disruptions (Caldara & Iacoviello, 2022). The I3E occupies an intermediate position, combining the responsiveness of market-based measures with a broader macroeconomic and international scope.

The empirical results further indicate that correlations across indices are generally modest, particularly between text-based and volatility-based measures, suggesting that they capture distinct dimensions of uncertainty (Ahir et al., 2022; Jurado et al., 2015). The decomposition of the I3E into financial, political, and geoeconomic components provides additional insight into these differences, revealing that each subindex aligns more closely with conceptually related measures such as the VIX, EPU, and GPR, respectively.

Our conditional-variance-based I3E measure of uncertainty, in the spirit of Jurado et al. (2015), seeks to isolate the conditional volatility of the unforecastable component of future outcomes, rather than relying on ex post realized volatility alone. The results show that this measure is highly correlated with benchmark macro uncertainty measures and exhibits stronger and statistically significant correlations with EPU and WUI than the I3E Global Spot index itself.

Overall, the findings suggest that no single measure fully captures the complexity of economic uncertainty. Instead, the choice of index should be guided by the research question at hand. The I3E index contributes to this landscape by offering a high-frequency, globally comparable measure that integrates multiple sources of uncertainty. In doing so, it provides a useful complement to existing indices, particularly in applications that require linking macroeconomic dynamics with financial, political, and geoeconomic developments in real time.

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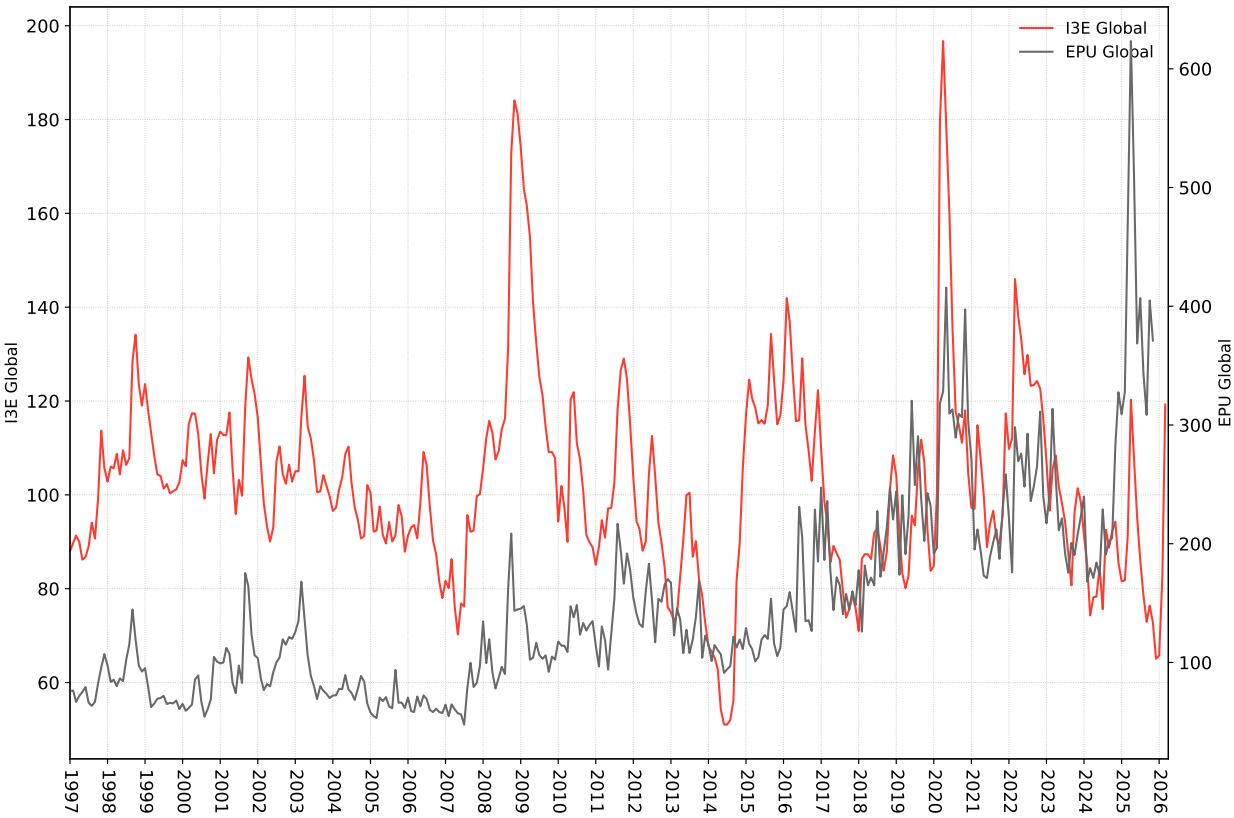

Figure 1: I3E vs EPU (global, monthly)

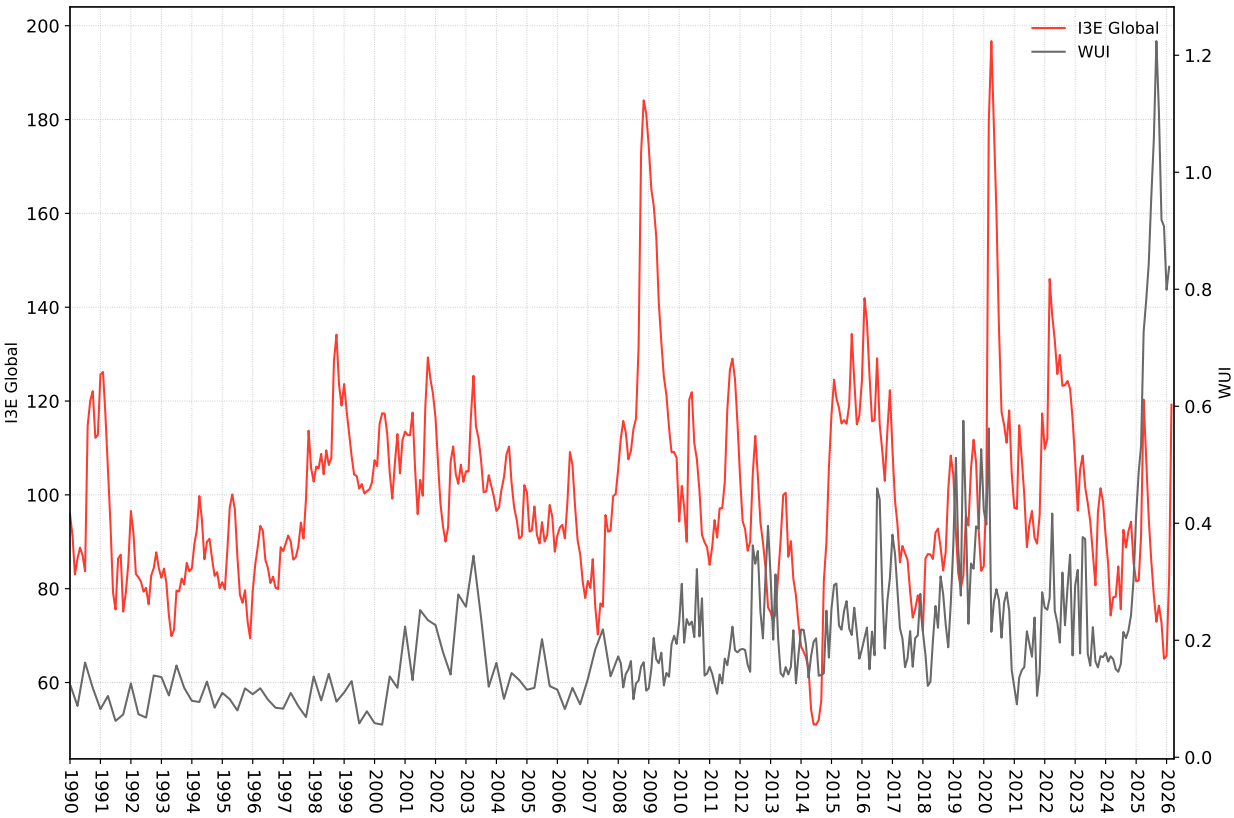

Figure 2: I3E vs WUI (global, monthly)

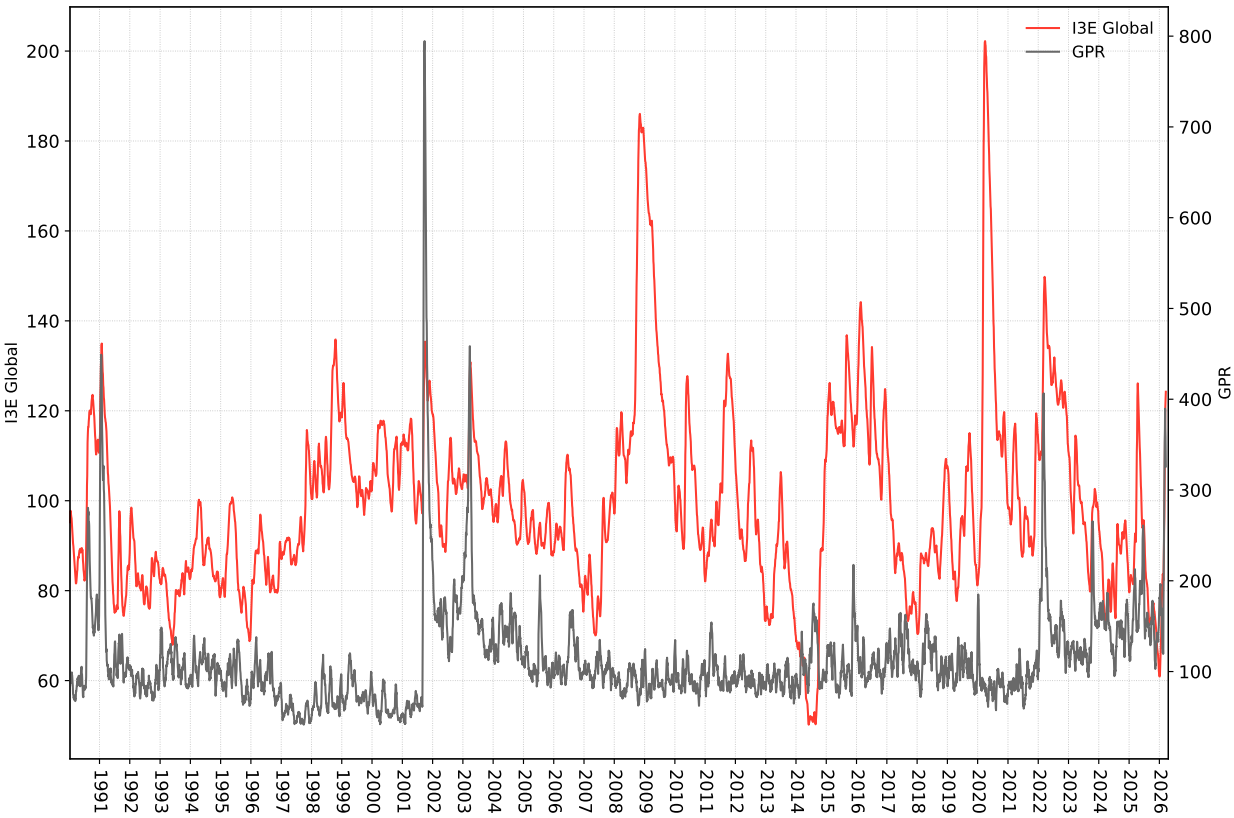

Figure 3: I3E vs GPR (global, daily)

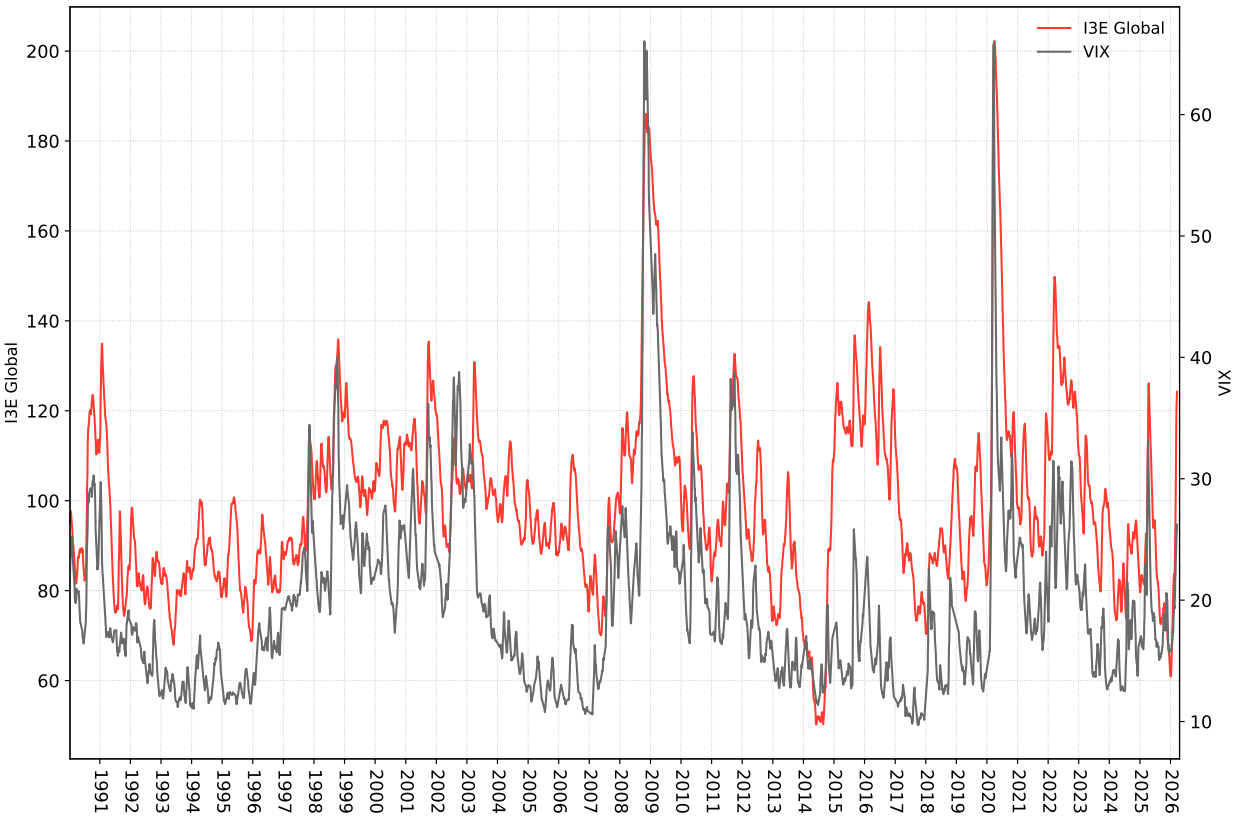

Figure 4: I3E vs VIX (global, daily)

Table 1: Macroeconomic Uncertainty Metrics

Metric	**Description**	**Reference**
Text-based		
Economic Policy Uncertainty (EPU)	Uncertainty keywords in newspapers	Baker et al. (2016)
World Uncertainty Index (WUI)	Frequency of uncertain EIU reports	Ahir et al. (2018)
Geopolitical Risk Index (GPR)	Newspaper-based measure of geopolitical tensions	Caldara (2022)
Volatility-based		
Volatility Index (VIX)	Implied volatility S&P 500 options	Whaley (1993)
Economic Uncertainty Index (I3E)	Volatility of stock market index, bonds, exchange rates, oil price	Garcia-Castro and Ariño (2026)
Stock Market Realized Volatility	High-frequency stock returns	Andersen et al. (2003)
Conditional Volatility (GARCH)	Time-varying volatility GARCH	Bollerslev (1986)
Macroeconomic Forecast Error	Deviation of outcomes from economic forecasts	Jurado et al. (2015)
Survey-based		
Forecaster Disagreement	Dispersion professional forecasts	Bomberger (1996)
Survey	Firm and household surveys	Bachmann et al. (2013)

Table 2: Correlations Among Macrouncertainty Measures

	EPU	WUI	GPR	VIX	I3E	$I3E_{Fin}$	$I3E_{Pol}$	$I3E_{Geo}$
EPU	1.000							
WUI	0.641	1.000						
GPR	0.178	0.223	1.000					
VIX	0.131	-0.035	0.091	1.000				
I3E	0.159	-0.076	0.104	0.760	1.000			
$I3E_{Fin}$	-0.056	-0.185	0.044	0.812	0.760	1.000		
$I3E_{Pol}$	0.543	0.121	0.012	0.262	0.517	0.050	1.000	
$I3E_{Geo}$	0.067	-0.049	0.148	0.657	0.946	0.626	0.382	1.000

Note: Variables are reported in their original scales and at monthly frequency. For comparability, daily series are converted to monthly frequency by taking monthly averages. Correlations with an absolute value greater than 0.10 are statistically significant at the 5 per cent level. For the WUI series, due to the smaller number of observations, the corresponding threshold is 0.11. Number of monthly observations: EPU: 347; WUI: 290; GPR: 435; VIX: 435; I3E: 435.

Table 3: Correlations (I3E Forward)

	EPU	WUI	GPR	VIX	I3E	$I3E_{F,US}$	$I3E_{F,EU}$	$I3E_{F,JP}$
EPU	1.000							
WUI	0.641	1.000						
GPR	0.178	0.223	1.000					
VIX	0.131	-0.035	0.091	1.000				
I3E	0.159	-0.076	0.104	0.760	1.000			
$I3E_{F,US}$	-0.023	-0.136	0.123	0.788	0.777	1.000		
$I3E_{F,EU}$	0.037	-0.132	0.123	0.747	0.793	0.955	1.000	
$I3E_{F,JP}$	0.054	0.060	0.071	0.559	0.607	0.781	0.743	1.000

Note: Variables are reported in their original scales and at monthly frequency. For comparability, daily series are converted to monthly frequency by taking monthly averages. Correlations with an absolute value greater than 0.10 are statistically significant at the 5 per cent level. For the WUI series, due to the smaller number of observations, the corresponding threshold is 0.11. Number of monthly observations: EPU: 347; WUI: 290; GPR: 435; VIX: 435; I3E: 435; $I3E_{F,US}$: 316; $I3E_{F,EU}$: 316; $I3E_{F,JP}$: 316.

Table 4: GARCH Model (Conditional Variance)

Variable	Mean equation (OLS)	Variance equation (GARCH)
Constant	13.039***	
	(0.652)	
$I3E_{F,USt}$	0.185***	
	(0.053)	
$I3E_{F,EUt}$	0.065	
	(0.052)	
$I3E_{F,JPt}$	0.072*	
	(0.037)	
$I3E_t$	1.352***	
	(0.081)	
$I3E_{t-1}$	-0.566***	
	(0.080)	
$I3E_{F,USt-1}$	0.000	
	(0.053)	
$I3E_{F,EUt-1}$	-0.120**	
	(0.052)	
$I3E_{F,JPt-1}$	-0.113***	
	(0.037)	
ω		0.050***
		(0.004)
α_1		0.693***
		(0.027)
β_1		0.229***
		(0.029)
$\alpha_1 + \beta_1$		0.922
Observations	6826	6826
R^2	0.765	
Adj. R^2	0.765	
Durbin–Watson	0.104	
Log-likelihood		-6218.21
AIC		12442.42
BIC		12462.90

Table 5: Correlations (Conditional Variance Uncertainty)

	EPU	WUI	GPR	VIX	I3E	U_{GCA}
EPU	1.000					
WUI	0.641	1.000				
GPR	0.178	0.223	1.000			
VIX	0.131	-0.035	0.091	1.000		
I3E	0.159	-0.076	0.104	0.760	1.000	
U_{GCA}	0.173	0.152	0.112	0.139	0.036	1.000

Note: Variables are reported in their original scales and at monthly frequency. For comparability, daily series are converted to monthly frequency by taking monthly averages. Correlations with an absolute value greater than 0.10 are statistically significant at the 5 per cent level. For the WUI series, due to the smaller number of observations, the corresponding threshold is 0.11. Number of monthly observations: EPU: 347; WUI: 290; GPR: 435; VIX: 435; I3E: 435; U_{GCA}: 315.